

10.6 MIDDLE EAST & AFRICA

The microbiome sequencing services market in the Middle East & Africa is projected to reach USD 5.55 million by 2028 from USD 3.55 million in 2023, at a CAGR of 9.3% from 2023 to 2028.

The increasing awareness and understanding of the importance of the microbiome in human health and diseases are contributing to the growth of this regional market. However, the market growth is likely to be restrained by factors such as the high costs associated with microbiome sequencing services and a lack of skilled professionals in the region. Despite these challenges, there are several opportunities for market growth in MEA, including the potential for increased investment in research and development, partnerships and collaborations between academic and industrial sectors, and the development of novel technologies and analytical tools.

TABLE 165 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Sequencing and Library Preparation	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Data Analysis	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 166 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Whole-genome Sequencing	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Shotgun Sequencing	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Other Types	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 167 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.33	1.46	1.60	1.75	1.91	2.10	2.30	2.52	9.5%
Sequencing by Ligation	0.38	0.42	0.46	0.50	0.54	0.59	0.65	0.71	9.2%
Nanopore Sequencing	0.15	0.18	0.20	0.24	0.27	0.31	0.36	0.41	15.2%
Other Technologies	1.10	1.19	1.29	1.39	1.50	1.63	1.76	1.91	8.1%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 168 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.93	1.04	1.15	1.28	1.43	1.59	1.77	1.97	11.3%
Academic and Research Institutes	1.35	1.47	1.59	1.73	1.87	2.03	2.21	2.41	8.6%
Other End Users	0.69	0.75	0.81	0.87	0.93	1.01	1.08	1.17	7.7%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.6.1 MIDDLE EAST & AFRICA: IMPACT OF RECESSION

The Middle East and Africa (MEA) region has experienced a significant economic slowdown in recent years, which has impacted the healthcare sector. The region has been hit by a combination of factors, including falling oil prices and political instability. The economic recession is having a severe impact on the healthcare industry, with healthcare spending reduced in many countries. Governments have been forced to reduce spending on healthcare due to budget constraints, and private healthcare providers have struggled to maintain profitability. This has led to a reduction in investment in healthcare infrastructure and a decline in the quality of care in some areas. However, there also are opportunities for growth in the MEA healthcare sector, particularly in the areas of digital health and telemedicine. The rising government initiatives to improve healthcare infrastructure and provide better access to advanced diagnostic tools in the region could also drive market growth. For instance, the UAE government has launched several initiatives to promote innovation in healthcare, including supporting research and development activities in the field of microbiome sequencing. These initiatives are expected to create a favorable environment for the growth of the microbiome sequencing services market in the region.